

Jeffrey William Shorthill

Salt Lake City, UT • 385-371-8083 • jeffrey.w.shorthill@gmail.com
jwshorthill.vercel.app • github.com/jeffreywilliamportfolio

SUMMARY

Progressive Insurance's first AI Prompt Engineer, now an independent AI researcher focused on interpretability, evaluation integrity, and equitable model behavior. Author of seven publicly archived studies on how language models route information, where evaluations mislead, and how safety behavior can be audited — including published self-corrections. Designed and built voice agents and AI tools for a global nonprofit focused on bringing technology to underserved populations.

SELECTED RESEARCH (INDEPENDENT, PUBLIC, REPRODUCIBLE)

- **In-Context Availability Reorganizes MoE Expert Routing** — cross-generation study (Qwen3.5 + 3.6-35B). Literal in-context availability shifts expert routing; the attention-retrieval mechanism is located and severed; the result replicates across a model generation. Implication: multi-turn evaluation contamination. doi:10.5281/zenodo.21892999
- **When Routing Entropy Tracks Length, Not Complexity** — self-refutation of a position confound in routing-entropy analyses; the “complexity gradient” reverses under correct measurement (DeepSeek V3.1, Qwen3.5-397B). doi:10.5281/zenodo.20779602
- **The Generation Half** — expert specialization is invisible in prefill and emerges during generation; rebuts a widely cited negative result; replicates at 122B. doi:10.5281/zenodo.20779604
- **No Single Safety Gate** — harm-refusal routing is distributed; suppressing the lead expert reallocates with zero harmful completions. doi:10.5281/zenodo.20786891
- **Dialect-Marked Response Audit** — matched-pair AAVE/AE safety-support equivalence; risk signal carried by register and action lexis. doi:10.5281/zenodo.20449546
- **Register-readout artifact bundle** (e114-artifacts v1.3.1) — preregistered sweeps that retired the program's earlier claims; raw router archives, frozen specs, and SHA-256 manifests public. doi:10.5281/zenodo.20709736
- **Methods** — preregistration, hash-frozen analysis rules, blinded and sham-controlled protocols, cross-runtime validation. Includes a fully blinded gravitational-wave echo search with positive-control gates.

EXPERIENCE

AI Consultant — Sole Proprietor (as Elevated Consultants LLC through 2026)

Feb 2024 – present

AI product consultancy; primary client: Gift Connect (nonprofit parenting program)

Salt Lake City, UT

- Shipped the Gift Connect Parenting Suite (iOS + Android, live on both stores) and six deployed AI tools: realtime voice agent (OpenAI Realtime API/WebRTC), song and story generators, Twilio SMS assistant with CDC milestone tools, GA4 analytics dashboard, cohort portal.
- Sole engineer across mobile (React Native/Expo), web (Next.js), backends (Express, Flask/Celery), and infra (Vercel, Heroku, Firebase, Postgres), with consent-gated analytics.

Progressive Insurance — AI Prompt Engineer (2023)

2018 – Dec 2023

Prior role: Customer Service Downtime Specialist (2018–2022)

- Proposed AI initiatives directly to executive leadership after GPT-4's release; a dedicated prompt-engineering role was created in response.
- Established internal prompting standards and taught applied AI sessions for data scientists, architects, and developers.

Webster Public Relations — Studio Manager, A/V Technical Specialist

2014 – 2018

Press-focused audio/video production

Nashville, TN

EDUCATION & SKILLS

Full Sail University — A.S., Recording Engineering (2014)

Research: experiment design, preregistration, bootstrap inference, routing/activation capture (custom llama.cpp + HF hooks), Jacobian-lens readouts, SAE probes, intervention studies • **Engineering:** Python, PyTorch, TypeScript, React/React Native, Swift/SwiftUI, Next.js, CUDA-cloud ops (Vast.ai), OpenAI/Anthropic APIs, agent frameworks